

Full Cost-Side Robustness Report

This memo consolidates the cost-side exchange-rate robustness evidence into one report. It is organized for co-author review rather than for direct journal submission. Two previously explored ideas are intentionally excluded from the final package: source-country fixed effects and lagged-*RER* timing-triplet specifications. The focus here is on the no-source-country-FE baseline, standalone and conditional forward-*RER* placebo designs, first-difference evidence, markup-channel diagnostics, and high-exposure visuals.

Executive Summary

The imported-parts cost channel remains visible across the core robustness checks, but the forward-*RER* placebo remains active. This remains true in the conditional current-plus-forward regression that controls for $\rho_{j,t-1} \log(RER_{jt})$ while testing $\rho_{j,t-1} \log(RER_{j,t+1})$. The cleanest interpretation is therefore narrower than the strongest version of the paper's current identification claim. The first-difference specifications and the foreign-assembled placebo are the most supportive pieces of evidence. The new decomposition results are designed to show whether the exchange-rate exposure term loads primarily on recovered marginal costs or on recovered markups.

Main Identification Table

Table 1: Main exchange-rate identification results

	Baseline current	Forward only	Current + forward	Foreign placebo
<i>Panel A: Levels</i>				
Current RER exposure term	0.598 (0.229)		-0.001 (0.214)	-0.049 (0.077)
Forward RER exposure term		0.739 (0.222)	0.740 (0.218)	
Observations	374	374	374	882
<i>Panel B: First differences</i>				
Current RER exposure term	0.445 (0.214)		0.213 (0.233)	-0.046 (0.080)
Forward RER exposure term		0.576 (0.212)	0.496 (0.229)	
Observations	255	255	255	590

Notes: Each cell reports the coefficient estimate with the clustered standard error below it. The baseline and foreign-placebo columns report single-regressor specifications; the current-plus-forward column reports the joint specification.

The main lessons from this table are simple. First, the foreign-assembled placebo stays near zero. Second, the forward-*RER* term remains active in both levels and first differences. Third, the

current-plus-forward specification should be treated as the main timing diagnostic because it controls directly for the contemporaneous exposure-weighted exchange-rate term. In that conditional specification, the future term remains large in levels and remains positive in first differences, while the current *RER* term looks materially stronger in first differences than in levels once the forward term is introduced.

Detailed Placebo Regressions

Table 2 reports the detailed levels placebo regressions. Table 3 reports the first-difference placebo regressions. In both tables, the current-plus-forward column is the conditional future-*RER* placebo: it tests whether the future exposure-weighted exchange-rate term predicts current recovered costs after controlling for the contemporaneous exposure-weighted exchange-rate term.

Table 2: Exchange-rate placebo regressions (levels)

Dependent Variable:	ln_costs			
	Domestic sample			Foreign placebo
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
ln(size)	0.5452*** (0.2007)	0.5037*** (0.1905)	0.5036*** (0.1909)	-0.1112 (0.1976)
ln(weight)	0.3021* (0.1749)	0.3843** (0.1597)	0.3844** (0.1620)	0.3867** (0.1549)
ln(hp)	0.1890** (0.0825)	0.1957** (0.0780)	0.1957** (0.0787)	0.2554*** (0.0532)
ln(mpg)	-0.0945 (0.0758)	-0.0657 (0.0779)	-0.0657 (0.0777)	-0.0930 (0.0651)
$\rho_{j,t-1} \times \log(RER_{jt})$	0.5980** (0.2292)		-0.0011 (0.2137)	-0.0495 (0.0772)
$\rho_{j,t-1} \times \log(RER_{j,t+1})$		0.7390*** (0.2222)	0.7398*** (0.2180)	
<i>Fixed-effects</i>				
make_model	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	374	374	374	882
R ²	0.97850	0.97929	0.97929	0.99152
Within R ²	0.18083	0.21088	0.21088	0.18728

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is log recovered marginal cost. Columns (1)–(3) use U.S.-assembled vehicles: column (1) includes the contemporaneous exchange-rate exposure term, column (2) replaces it with the one-year-ahead term, and column (3) includes both terms. Column (4) is a foreign-assembled placebo estimated with the contemporaneous exposure term on a different sample; it tests whether the same exposure construction predicts costs where the domestic imported-parts channel should not apply. All specifications include make-model and year fixed effects and vehicle controls. Standard errors are clustered by make-model.

Table 3: Exchange-rate placebo regressions (first differences)

Dependent Variable:	ln_costs			
Model:	Domestic sample			Foreign placebo
	(1)	(2)	(3)	(4)
<i>Variables</i>				
$\Delta \ln(\text{size})$	0.5088*	0.5373**	0.5250**	-0.0119
	(0.2569)	(0.2563)	(0.2550)	(0.1926)
$\Delta \ln(\text{weight})$	0.4020*	0.4201*	0.4077*	0.2900**
	(0.2222)	(0.2225)	(0.2231)	(0.1303)
$\Delta \ln(\text{hp})$	0.2137**	0.2065**	0.2061**	0.2855***
	(0.0998)	(0.0968)	(0.0978)	(0.0569)
$\Delta \ln(\text{mpg})$	-0.0448	-0.0565	-0.0559	-0.1342*
	(0.0692)	(0.0758)	(0.0742)	(0.0787)
$\rho_{j,t-1} \times \Delta \log(RER_{jt})$	0.4452**		0.2133	-0.0457
	(0.2136)		(0.2329)	(0.0805)
$\rho_{j,t-1} \times \Delta \log(RER_{j,t+1})$		0.5756***	0.4958**	
		(0.2120)	(0.2286)	
<i>Fixed-effects</i>				
year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	255	255	255	590
R ²	0.14428	0.15335	0.15492	0.32981
Within R ²	0.11390	0.12329	0.12492	0.23399

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the first difference of log recovered marginal cost. Columns (1)–(3) use U.S.-assembled vehicles observed in consecutive years: column (1) includes the contemporaneous exchange-rate exposure term, column (2) replaces it with the one-year-ahead exchange-rate difference, and column (3) includes both terms. Column (4) is a foreign-assembled placebo estimated on a different first-difference sample. All specifications include year fixed effects and differenced vehicle controls. Standard errors are clustered by make-model.

Sample and Specification Robustness

Table 4 reports the canonical pass-through coefficient in levels and first differences. Table 5 adds an unrestricted exchange-rate main effect. Table 6 replaces the baseline exposure measure with alternative exposure definitions in the preferred first-difference designs. Table 7 summarizes the leave-one-country-out checks.

Table 4: Baseline pass-through estimates

Dependent Variable:	ln_costs	
Model:	(1)	(2)
<i>Variables</i>		
ln(size)	0.5452*** (0.2007)	0.5088* (0.2569)
ln(weight)	0.3021* (0.1749)	0.4020* (0.2222)
ln(hp)	0.1890** (0.0825)	0.2137** (0.0998)
ln(mpg)	-0.0945 (0.0758)	-0.0448 (0.0692)
$\rho_{j,t-1} \times \log(RER_{jt})$	0.5980** (0.2292)	0.4452** (0.2136)
<i>Fixed-effects</i>		
make_model	Yes	
year	Yes	Yes
<i>Fit statistics</i>		
Observations	374	255
R ²	0.97850	0.14428
Within R ²	0.18083	0.11390

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is log recovered marginal cost. Column (1) is the levels specification with make-model and year fixed effects. Column (2) is the first-difference specification with year fixed effects; the smaller sample reflects the requirement that a make-model be observed in consecutive years with non-missing differenced costs, exchange rates, exposure, and controls. The exchange-rate term is interacted with lagged imported-parts exposure. Vehicle controls are log size, log weight, log horsepower, and log miles per gallon. Standard errors are clustered by make-model.

Table 5: Pass-through robustness with an exchange-rate main effect

Dependent Variable:	ln_costs	
Model:	(1)	(2)
<i>Variables</i>		
$\log(RER_{jt})$	-0.2212*	-0.2212
	(0.1199)	(0.2064)
$\ln(\text{size})$	0.5406**	0.4972*
	(0.2060)	(0.2700)
$\ln(\text{weight})$	0.3354*	0.3904*
	(0.1695)	(0.2316)
$\ln(\text{hp})$	0.1959**	0.2205**
	(0.0848)	(0.1008)
$\ln(\text{mpg})$	-0.0808	-0.0406
	(0.0767)	(0.0682)
$\rho_{j,t-1} \times \log(RER_{jt})$	1.230***	1.148*
	(0.4175)	(0.6587)
<i>Fixed-effects</i>		
make_model	Yes	
year	Yes	Yes
<i>Fit statistics</i>		
Observations	374	255
R ²	0.97882	0.14802
Within R ²	0.19302	0.11776

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is log recovered marginal cost. Column (1) is the levels specification with make-model and year fixed effects. Column (2) is the first-difference specification with year fixed effects. Both columns add an unrestricted source-country exchange-rate term to the exposure interaction. Vehicle controls are log size, log weight, log horsepower, and log miles per gallon. Standard errors are clustered by make-model.

Elasticity-Interaction Robustness

Table 8 re-estimates the elasticity-interaction specification on the canonical domestic sample. The point of this table is not to settle identification on its own, but to document the heterogeneity pattern used by optional counterfactual paths.

Markup-Channel Diagnostics

Table 9 decomposes the baseline current-*RER* result across recovered marginal costs, observed prices, and recovered markups. Table 10 repeats the same exercise for the forward-*RER* placebo. If the exchange-rate exposure term were showing up mainly in recovered markups rather than in recovered costs, that would be a direct warning sign for the paper’s current interpretation of η .

Table 6: Alternative exposure definitions in first-difference cost regressions

Dependent Variable:	ln_costs					
	Current RER		Forward RER			
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
$\Delta \ln(\text{size})$	0.5088*	0.5095*	0.5051*	0.5373**	0.5384**	0.5370**
	(0.2569)	(0.2573)	(0.2653)	(0.2563)	(0.2568)	(0.2544)
$\Delta \ln(\text{weight})$	0.4020*	0.4067*	0.4003*	0.4201*	0.4120*	0.4533**
	(0.2222)	(0.2231)	(0.2232)	(0.2225)	(0.2218)	(0.2107)
$\Delta \ln(\text{hp})$	0.2137**	0.2148**	0.2246**	0.2065**	0.2050**	0.2066**
	(0.0998)	(0.1000)	(0.0998)	(0.0968)	(0.0963)	(0.0951)
$\Delta \ln(\text{mpg})$	-0.0448	-0.0464	-0.0344	-0.0565	-0.0553	-0.0602
	(0.0692)	(0.0699)	(0.0665)	(0.0758)	(0.0754)	(0.0772)
Primary-share exposure term	0.4452**			0.5756***		
	(0.2136)			(0.2120)		
Observed-total exposure term		0.4221*			0.5841***	
		(0.2145)			(0.2057)	
$\Delta \log(RER_{jt})$			0.0291			
			(0.0728)			
Top-quartile exposure interaction			0.3725*			-0.1161
			(0.1888)			(0.1176)
$\Delta \log(RER_{j,t+1})$						0.2645***
						(0.0814)
<i>Fixed-effects</i>						
year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	255	255	255	255	255	255
R ²	0.14428	0.14369	0.15317	0.15335	0.15476	0.16736
Within R ²	0.11390	0.11328	0.12310	0.12329	0.12475	0.13780

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is the first difference of log recovered marginal cost. Columns (1)–(3) use contemporaneous exchange-rate differences; columns (4)–(6) use one-year-ahead exchange-rate differences. Columns (1) and (4) use the lagged primary-source foreign-parts share. Columns (2) and (5) use the lagged sum of the first two observed foreign-parts shares. Columns (3) and (6) use an indicator for lagged primary-source exposure in the top quartile and include the unrestricted exchange-rate main effect. All specifications include year fixed effects and differenced vehicle controls. Standard errors are clustered by make-model.

Table 7: Leave-one-country-out baseline pass-through regressions

Omitted country	Levels coef.	Levels N	FD coef.	FD N
Full sample	0.598 (0.229)	374	0.445 (0.214)	255
Mexico	1.250 (0.496)	199	1.117 (0.497)	143
Japan	0.605 (0.512)	227	-0.027 (0.451)	142
Korea	0.628 (0.229)	341	0.490 (0.222)	238
Germany	0.608 (0.224)	357	0.459 (0.218)	243
Spain	0.597 (0.229)	372	0.435 (0.215)	254

Notes: Each row re-estimates the baseline contemporaneous exchange-rate pass-through regression after omitting observations whose primary source country is listed in the first column. The full-sample row imposes no country omission. The levels specification includes make-model and year fixed effects and vehicle controls. The first-difference specification uses consecutive-year differences and year fixed effects. Coefficients are reported with make-model clustered standard errors in parentheses.

Table 8: Elasticity-interaction diagnostics

Dependent Variable:	ln_costs	
Model:	(1)	(2)
<i>Variables</i>		
ln(size)	0.4895** (0.1941)	0.4983** (0.2475)
ln(weight)	0.2777 (0.1710)	0.4015* (0.2213)
ln(hp)	0.2028** (0.0786)	0.2034** (0.0963)
ln(mpg)	-0.0720 (0.0727)	-0.0276 (0.0698)
$\rho_{j,t-1} \times \log(RER_{jt})$	4.511*** (1.336)	3.936* (2.075)
$\rho_{j,t-1} \times \log \varepsilon_{j,t-1} \times \log(RER_{jt})$	-2.031*** (0.6799)	-1.783* (1.063)
<i>Fixed-effects</i>		
make_model	Yes	
year	Yes	Yes
<i>Fit statistics</i>		
Observations	374	255
R ²	0.97961	0.15817
Within R ²	0.22295	0.12828

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is log recovered marginal cost. Column (1) is the levels specification with make-model and year fixed effects. Column (2) is the first-difference specification with year fixed effects. The table augments the baseline exposure interaction with an interaction between lagged imported-parts exposure, the exchange-rate term, and lagged log absolute own-price elasticity. Standard errors are clustered by make-model.

Table 9: Baseline exchange-rate decomposition across recovered costs, prices, and markups

Dependent Variables:	ln_costs	ln_prices	ln_markups	y		
		Levels		First differences		
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(size)	0.5452*** (0.2007)	0.4176** (0.1672)	-0.5763*** (0.1731)	0.5088* (0.2569)	0.4328* (0.2233)	-0.3552* (0.1918)
ln(weight)	0.3021* (0.1749)	0.2432 (0.1509)	-0.3766** (0.1874)	0.4020* (0.2222)	0.3315 (0.2084)	-0.4175** (0.2004)
ln(hp)	0.1890** (0.0825)	0.1611** (0.0739)	-0.1720*** (0.0605)	0.2137** (0.0998)	0.1736** (0.0869)	-0.2575*** (0.0862)
ln(mpg)	-0.0945 (0.0758)	-0.0896 (0.0629)	0.0670 (0.0740)	-0.0448 (0.0692)	-0.0631 (0.0571)	-0.0833 (0.0815)
Current RER exposure term	0.5980** (0.2292)	0.4829** (0.1901)	-0.3523 (0.2352)	0.4452** (0.2136)	0.3625** (0.1702)	-0.3121 (0.2990)
<i>Fixed-effects</i>						
make_model	Yes	Yes	Yes			
year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	374	374	374	255	255	255
R ²	0.97850	0.97931	0.96766	0.14428	0.14463	0.15772
Within R ²	0.18083	0.16751	0.17948	0.11390	0.10498	0.14781

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Columns (1)–(3) are levels regressions with make-model and year fixed effects; columns (4)–(6) are first-difference regressions with year fixed effects. The dependent variables are log recovered marginal cost, log observed price, and log recovered markup, respectively. The reported exchange-rate coefficient is the contemporaneous source-country exchange-rate term interacted with lagged imported-parts exposure. All regressions include the vehicle controls shown in the table. Standard errors are clustered by make-model.

Table 10: Forward-RER placebo decomposition across recovered costs, prices, and markups

Dependent Variables:	ln_costs	ln_prices	ln_markups	y		
		Levels		First differences		
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
ln(size)	0.5037*** (0.1905)	0.3857** (0.1601)	-0.5371*** (0.1604)	0.5373** (0.2563)	0.4561** (0.2228)	-0.3747* (0.1915)
ln(weight)	0.3843** (0.1597)	0.3071** (0.1376)	-0.4470** (0.1825)	0.4201* (0.2225)	0.3459* (0.2069)	-0.4315** (0.1994)
ln(hp)	0.1957** (0.0780)	0.1666** (0.0701)	-0.1746*** (0.0589)	0.2065** (0.0968)	0.1674* (0.0843)	-0.2538*** (0.0848)
ln(mpg)	-0.0657 (0.0779)	-0.0673 (0.0648)	0.0416 (0.0776)	-0.0565 (0.0758)	-0.0732 (0.0626)	-0.0768 (0.0843)
Forward RER exposure term	0.7390*** (0.2222)	0.5822*** (0.1867)	-0.5635*** (0.2089)	0.5756*** (0.2120)	0.4881*** (0.1777)	-0.3372 (0.2262)
<i>Fixed-effects</i>						
make_model	Yes	Yes	Yes			
year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	374	374	374	255	255	255
R ²	0.97929	0.97989	0.96897	0.15335	0.15366	0.16016
Within R ²	0.21088	0.19112	0.21269	0.12329	0.11443	0.15028

Clustered (make_model) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Columns (1)–(3) are levels regressions with make-model and year fixed effects; columns (4)–(6) are first-difference regressions with year fixed effects. The dependent variables are log recovered marginal cost, log observed price, and log recovered markup, respectively. The reported exchange-rate coefficient is the one-year-ahead source-country exchange-rate term interacted with lagged imported-parts exposure. All regressions include the vehicle controls shown in the table. Standard errors are clustered by make-model.

High-Exposure Visual Evidence

Figure 1 plots indexed year-level medians of recovered marginal cost, observed price, and the primary-source *RER* for the high-exposure domestic vehicles only. Figure 2 then shows residualized first-difference patterns for recovered marginal costs, observed prices, and recovered markups against residualized exchange-rate changes, again on the high-exposure sample only.

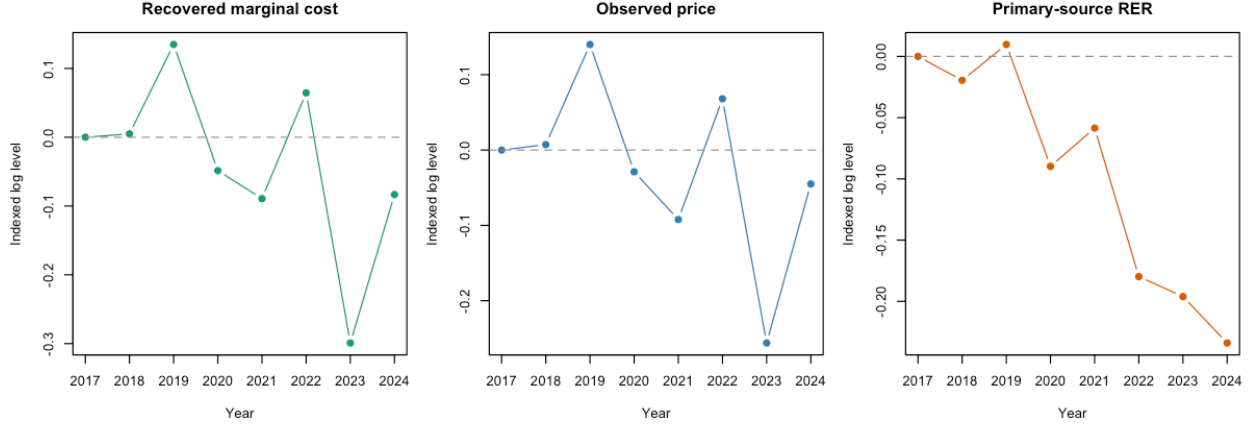


Figure 1: Indexed year-series for high-exposure domestic vehicles

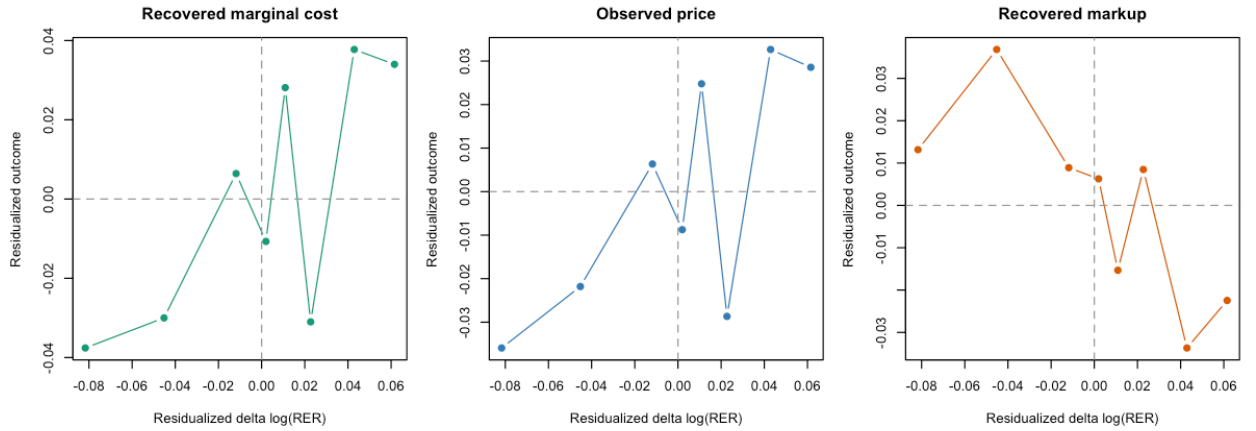


Figure 2: Residualized first-difference patterns for high-exposure domestic vehicles

Lessons From Related Papers

The related exchange-rate pass-through literature suggests three practical lessons for our empirical design.

First, the right benchmark object is heterogeneous pass-through rather than an average exchange-rate coefficient. This is clearest in Amiti, Itskhoki, and Konings, where import intensity proxies marginal-cost sensitivity and market share proxies markup elasticity. That reinforces the choice to keep the exposure interaction central.

Second, the more informative robustness checks are about decomposition and differential responses, not about adding a standalone source-country fixed effect. The structural papers by Goldberg and Verboven and by Goldberg and Hellerstein emphasize separating traded-cost responses from local-cost or markup adjustment. The price/markup decomposition in this memo is the closest analogue we can implement in the current repo.

Third, the literature is comfortable with rich time and market controls, but it does not suggest that a time-invariant source-country effect should be the main credibility test in a product fixed-effect design. That is why the current memo drops the source-country-FE branch and instead emphasizes placebo timing, sample tightening, alternative exposure definitions, and decomposition.

Bottom Line

The robustness package continues to support the existence of an imported-parts cost channel, especially in first differences and relative to the foreign-assembled placebo. The canonical panel now excludes product-years with conflicting raw primary source countries while allowing share averaging when the primary country agrees. At the same time, the forward-*RER* placebo remains active, so the cleanest interpretation is still narrower than a fully resolved causal timing design. The report therefore points toward a practical revision strategy: keep the channel, emphasize the first-difference evidence, document the remaining timing concern clearly, and avoid claiming that the current placebo evidence fully settles the identification question.